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NEUROINSIGHT AI RADIOLOGY REPORT

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Scan ID   : MRI-2026-66911

Date/Time : 27 May 2026 | 03:16 PM

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1. MRI VALIDATION

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Result:

Valid Brain MRI Scan Detected

Validation Confidence:

99.99%

Observation:

The uploaded image was successfully identified as a brain MRI scan and passed all validation checks for further tumor analysis.

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2. TUMOR CLASSIFICATION

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Detection Result:

Tumor Detected

Tumor Type:

Meningioma

Model Confidence:

90.75%

Classification Model:

Deep Learning CNN (EfficientNet-based)

AI Interpretation:

The deep learning model predicts the presence of a Meningioma with high confidence.

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3. TUMOR SEGMENTATION ANALYSIS

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Affected Brain Region:

0.82%

Tumor Area:

536 pixels

Severity Level:

Very Small

Tumor Location:

X = 78.7, Y = 85.0

Bounding Box Size:

[28, 26]

SHAPE CHARACTERISTICS

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1. ECCENTRICITY = 0.614

Scale: 0.0 (perfect circle) -> 1.0 (elongated line)

Shape Description:

Moderately elongated

Interpretation:

Elevated eccentricity - possible infiltrative growth pattern

Clinical Significance:

Noticeably irregular, non-circular form. May suggest infiltrative characteristics or pressure-induced deformation. Associated with higher-grade pathology in some tumor types.

2. SOLIDITY = 0.947

Scale: 0.0 (highly irregular) -> 1.0 (perfect solid shape)

Shape Description:

Compact and convex

Interpretation:

High solidity - well-defined, likely benign margin

Clinical Significance:

Highly solid, well-defined boundary with no significant concavities. Strongly associated with benign, encapsulated lesions. Tumor has a clear, predictable margin - favorable for surgical planning.

Combined Shape Risk Flag:

LOW SHAPE RISK - Compact, regular boundaries consistent with benign morphology.

Segmentation Observation:

The segmentation model identified the suspected tumor

region and highlighted the affected area in the MRI.

The detected region appears compact and convex

with compact boundary characteristics.

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4. TUMOR TYPE - CLINICAL PROFILE

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Tumor Type:

Meningioma

Description:

Meningioma arises from the meninges - the protective membranes surrounding the brain and spinal cord. It is typically slow-growing, extra-axial, and often benign, representing the most common primary brain tumor (~36%). Most meningiomas are incidental findings on imaging.

Key Characteristics:

- Slow-growing
- Extra-axial (outside brain tissue)
- Well-defined margins
- Dural tail sign on MRI
- Often asymptomatic when small

Origin / Cell Type:

Arachnoid cap cells of the meninges

Prevalence:

~36% of all primary brain tumors

Typical Brain Location:

Convexity, falx cerebri, sphenoid wing, posterior fossa

Growth Rate:

Slow (average 1-2 mm/year)

Malignancy Profile:

Usually benign - WHO Grade I (occasionally II or III)

Common Symptoms:

Headache, seizures, focal neurological deficits (if large)

Treatment Options:

Observation / Microsurgery / Stereotactic radiosurgery

Prognosis:

Generally favorable; recurrence possible after resection

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5. GRAD-CAM EXPLAINABILITY ANALYSIS

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Explainable AI (Grad-CAM):

The Grad-CAM heatmap highlights the regions of the MRI

image that contributed most strongly to the prediction.

Observation:

The highlighted heat regions correspond closely with the segmented tumor area, improving prediction i nterpretability and trustworthiness. The activation map confirms the model focused on the correct anat omical region.

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6. RISK ASSESSMENT

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Risk Level:

LOW RISK

Reason:

The detected tumor occupies ~0.82% of the brain region (Very Small classification).

Shape-Based Morphology Risk:

LOW SHAPE RISK - Compact, regular boundaries consistent with benign morphology.

Suggested Medical Action:

Regular monitoring recommended. Neurologist consultation advised. Follow-up MRI in 6-12 months.

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7. AI GENERATED MEDICAL SUMMARY

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The MRI indicates a Very Small Meningioma occupying ~0.82% of the brain region. The lesion appear s localized and well-defined with no evidence of aggressive infiltration. Low-risk profile warrants regular monitoring rather than immediate surgical intervention.

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8. RECOMMENDATIONS

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Recommended Next Steps:

1. Consult a neurologist or neurosurgeon for formal evaluation
2. Periodic MRI surveillance every 6-12 months
3. No immediate surgical intervention for very small (<3 cm) tumors
4. Monitor for neurological symptom onset or progression
5. Consider stereotactic radiosurgery (SRS) if growth is detected

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9. IMPORTANT DISCLAIMER

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This report is generated using Artificial Intelligence
and deep learning models for educational and research
purposes only.

The system assists medical professionals by providing
automated MRI analysis and should NOT be considered a
final clinical diagnosis.

Final diagnosis must always be confirmed by qualified medical experts including radiologists, neurologists, and neurosurgeons.

Eccentricity and Solidity values are shape descriptors derived from image segmentation. They are supportive indicators only and do not replace histopathological confirmation.

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END OF AI GENERATED REPORT

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